

Performance Testing

Date	15 July 2026
Team ID	XXXXX
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and reliability of the FinRelief AI application under normal and peak workloads. The objective was to ensure that the application can efficiently handle multiple users, process financial data, and generate AI-powered debt settlement recommendations without significant delays.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, Postman
Type of Testing	Load Testing, Stress Testing, API Performance Testing
Target Module / API	User Authentication, Loan Management, Financial Analysis, AI Recommendation APIs
Test Environment	Local Development Environment (React + FastAPI + SQLite)
Test Date	11 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login and Authentication API	50 Users	60 sec	Login requests processed successfully with response time below 2 seconds.
2	Loan Management (Add, Update, Delete Loans)	75 Users	90 sec	Loan operations completed successfully without database errors.
3	Financial Health Analysis API	100 Users	120 sec	Financial analysis generated accurately with stable performance.
4	AI Debt Settlement Recommendation	50 Users	120 sec	AI recommendations generated successfully with acceptable response time.

Performance Test Summary

Metric	Result
Total Test Scenarios	4
APIs Tested	4
Average Response Time	< 2 seconds
Maximum Concurrent Users	100
Test Status	Passed
Application Stability	Stable under expected workload